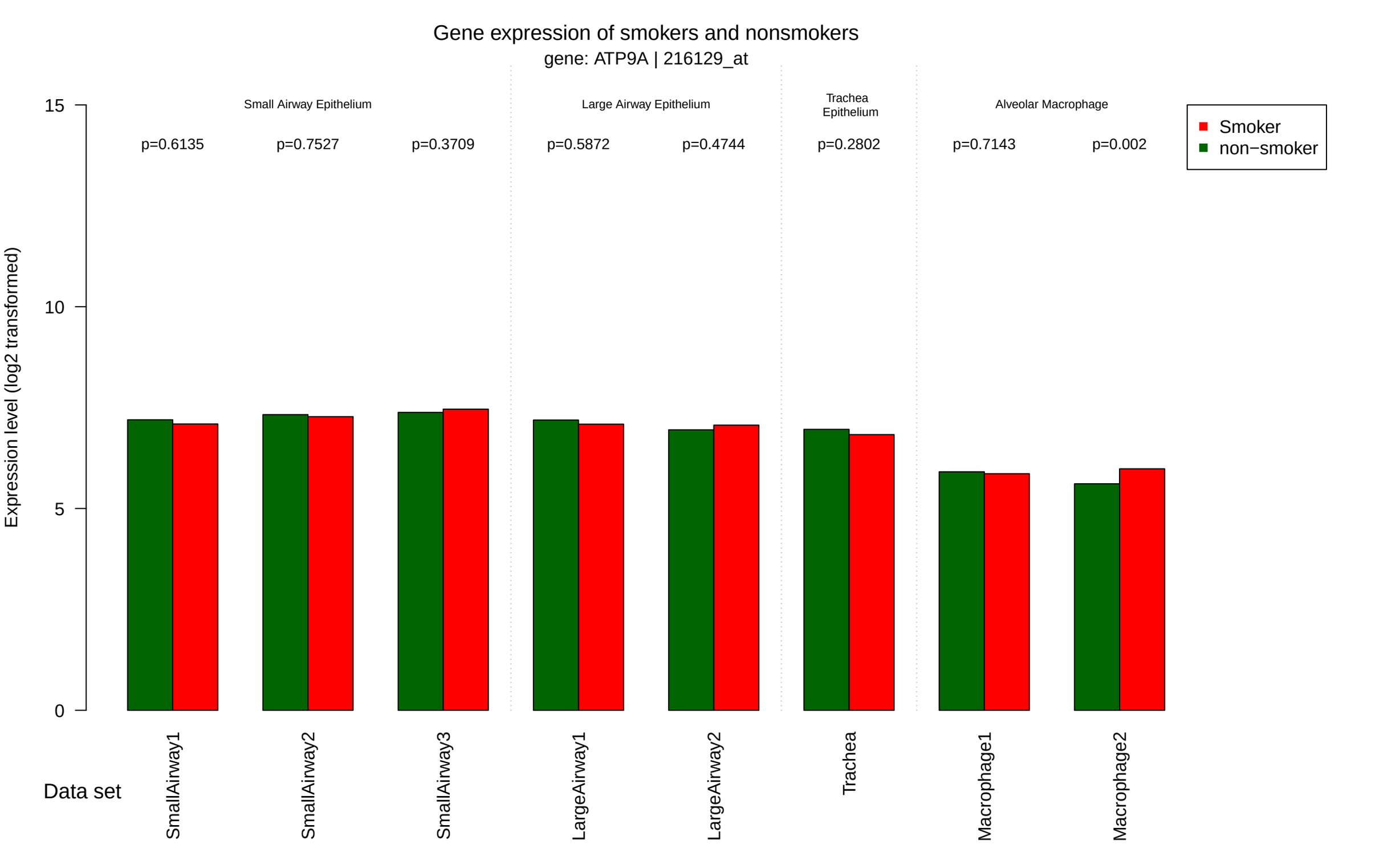
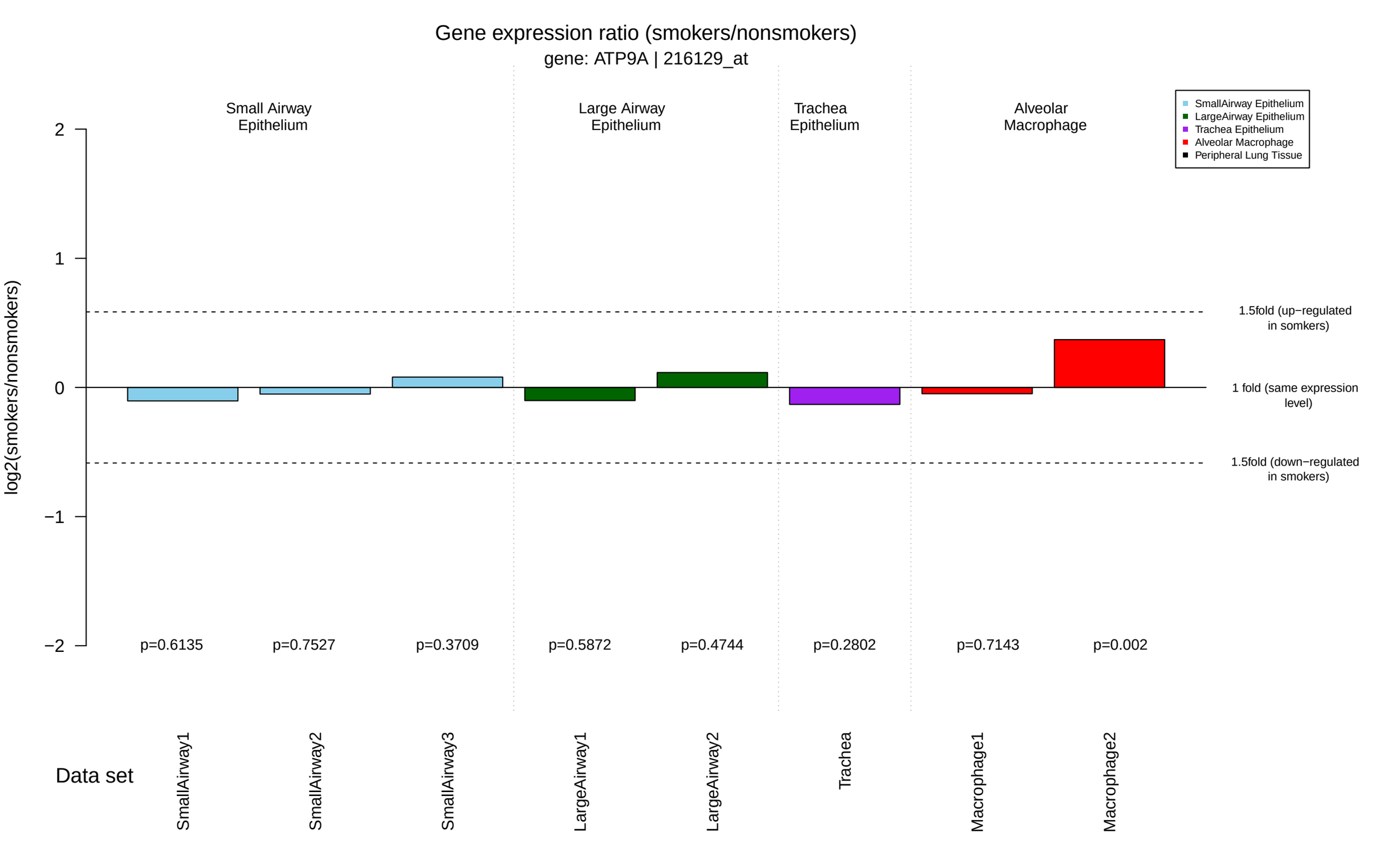
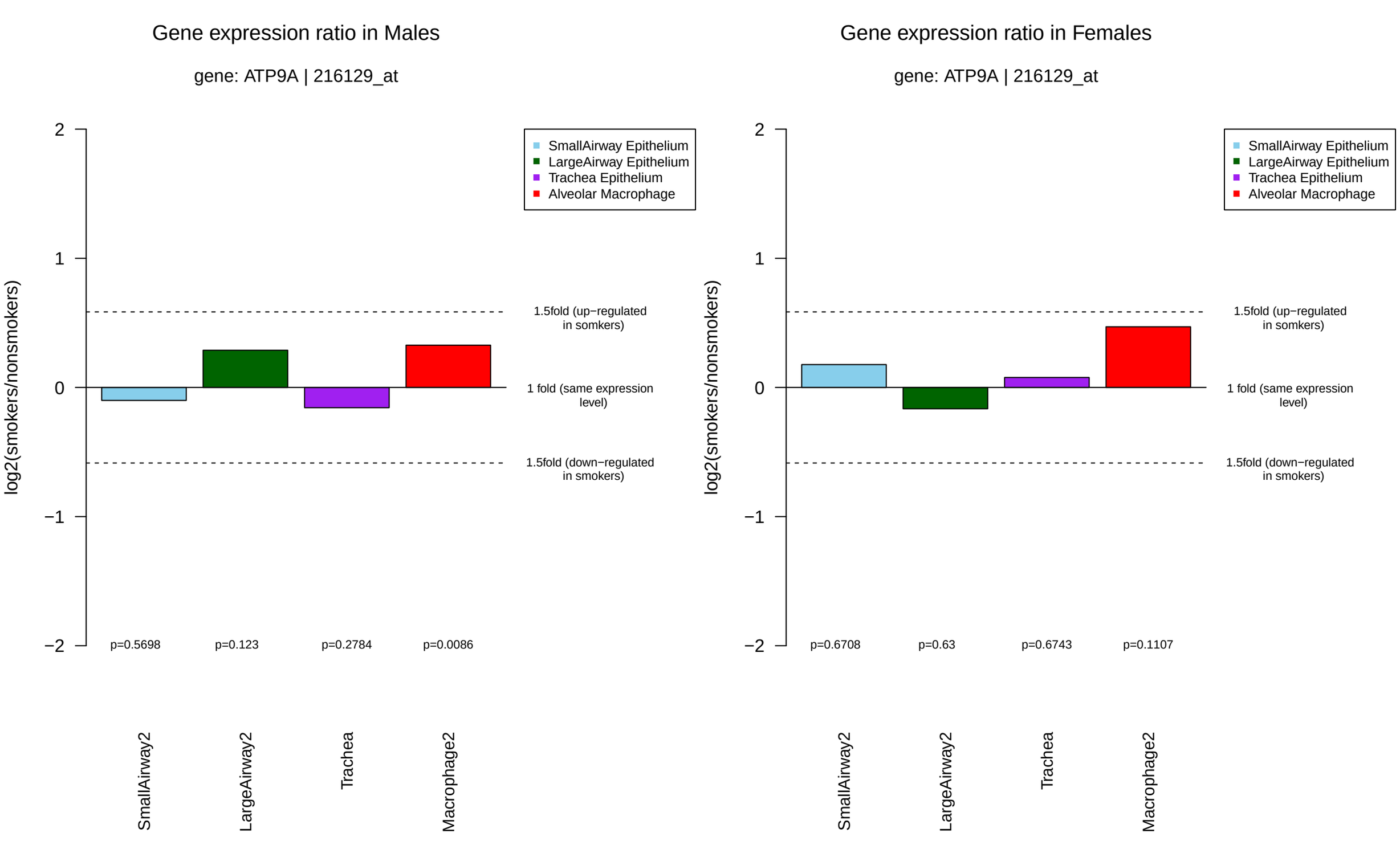
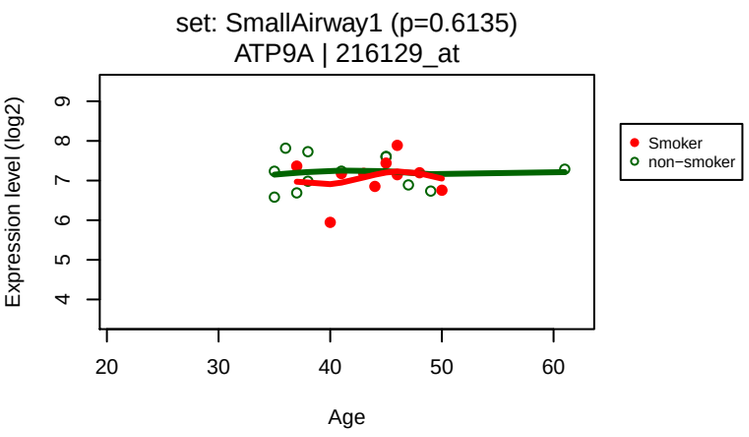
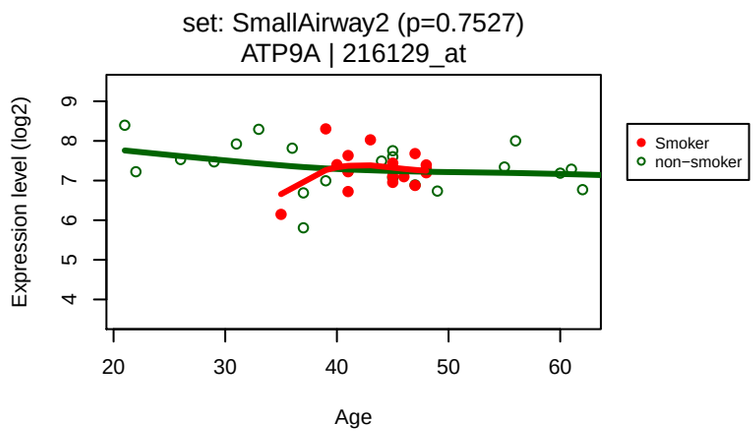
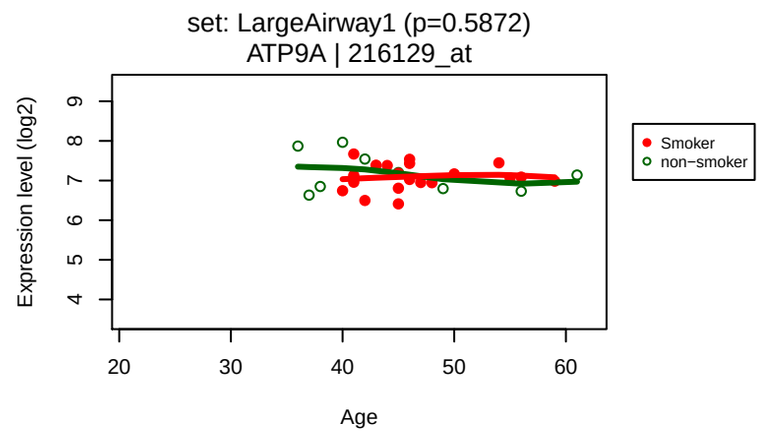

Gene expression of smokers and nonsmokers
gene: ATP9A | 216129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6135
p=0.7527
p=0.3709
p=0.5872
p=0.4744
p=0.2802
p=0.7143
p=0.002
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

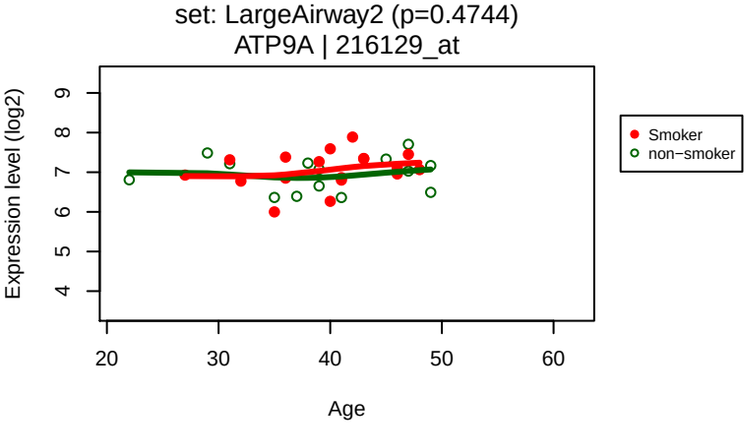

Gene expression ratio (smokers/nonsmokers)
gene: ATP9A | 216129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6135
p=0.7527
p=0.3709
p=0.5872
p=0.4744
p=0.2802
p=0.7143
p=0.002
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

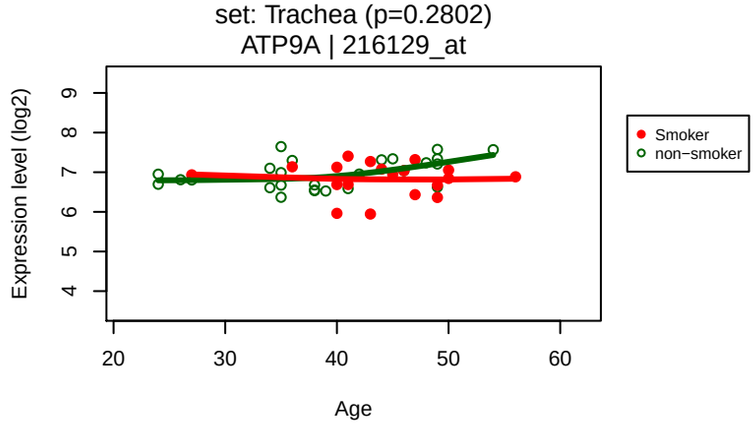

Gene expression ratio in Males
gene: ATP9A | 216129_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5698
p=0.123
p=0.2784
p=0.0086
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP9A | 216129_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6708
p=0.63
p=0.6743
p=0.1107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

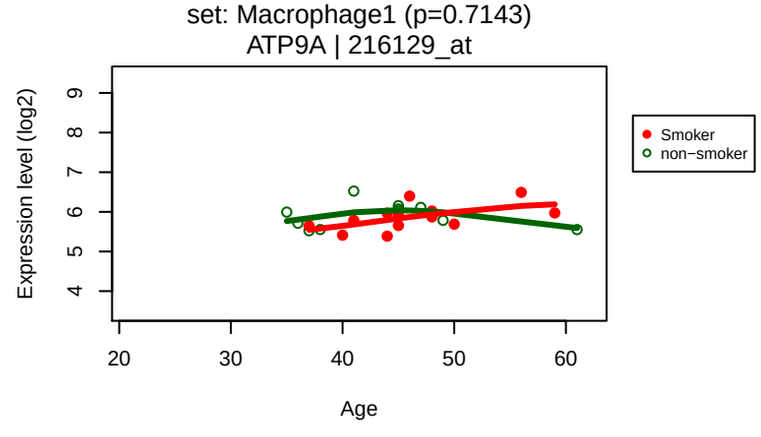

set: SmallAirway1 (p=0.6135)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker

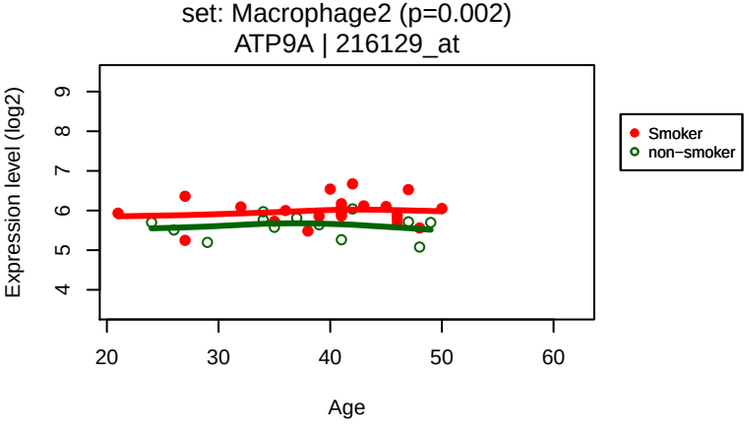

set: SmallAirway2 (p=0.7527)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker

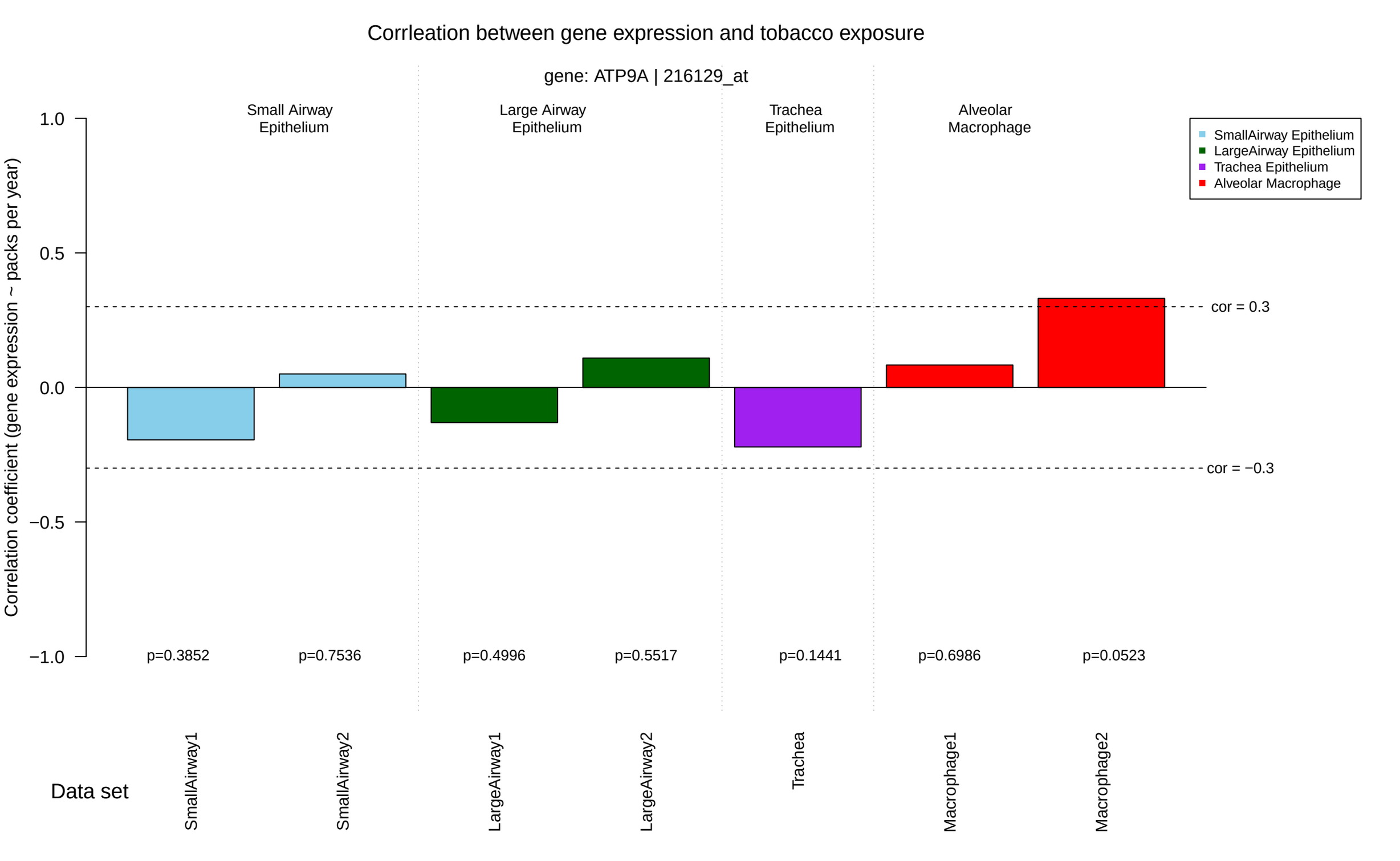

set: LargeAirway1 (p=0.5872)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4744)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2802)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7143)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.002)
ATP9A | 216129_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP9A | 216129_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3852
p=0.7536
p=0.4996
p=0.5517
p=0.1441
p=0.6986
p=0.0523
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2